

lab7_li

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(DBI)
library(RSQLite)
conn <- dbConnect(SQLite(), "database.sqlite3")
```

```
dbListTables(conn)
```

```
[1] "course_offerings"      "courses"              "grade_distributions"
[4] "instructors"           "rooms"                 "schedules"
[7] "sections"             "subject_memberships"  "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT');

```
SELECT DISTINCT i.name FROM SUBJECTS AS s
INNER JOIN subject_memberships AS sm ON sm.subject_code = s.code
INNER JOIN sections AS sec ON sec.course_offering_uuid = sm.course_offering_uuid
INNER JOIN teachings AS t ON t.section_uuid = sec.uuid
INNER JOIN instructors AS i ON i.id = t.instructor_id
WHERE s.abbreviation = 'STAT'
```

Table 1: Displaying records 1 - 10

name
MINJING TAO
DONALD PORTER
SHENG WANG

name
KUNLING HUANG
DONG XIA
JENNIFER NGUYEN
SEULKEE YUN
DUZHE WANG
YONGFENG WU
JARED BROWN

```
SELECT COUNT(DISTINCT i.id) AS how_many_stat_instructors FROM SUBJECTS AS s
INNER JOIN subject_memberships AS sm ON sm.subject_code = s.code
INNER JOIN sections AS sec ON sec.course_offering_uuid = sm.course_offering_uuid
INNER JOIN teachings AS t ON t.section_uuid = sec.uuid
INNER JOIN instructors AS i ON i.id = t.instructor_id
WHERE s.abbreviation = 'STAT'
```

Table 2: 1 records

how_many_stat_instructors
379

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Quem é o professor mais difícil?

```
SELECT i.name FROM instructors AS i
INNER JOIN teachings AS t ON t.instructor_id = i.id
INNER JOIN sections AS s ON t.section_uuid = s.uuid
INNER JOIN course_offerings AS co ON co.uuid = s.course_offering_uuid
INNER JOIN grade_distributions AS gd ON co.uuid = gd.course_offering_uuid
GROUP BY i.name
ORDER BY SUM(gd.f_count) DESC
LIMIT 1
```

Table 3: 1 records

name
DEBRA DEPPELER

Quem é o professor mais fácil?

```
SELECT i.name FROM instructors AS i
INNER JOIN teachings AS t ON t.instructor_id = i.id
INNER JOIN sections AS s ON t.section_uuid = s.uuid
INNER JOIN course_offerings AS co ON co.uuid = s.course_offering_uuid
INNER JOIN grade_distributions AS gd ON co.uuid = gd.course_offering_uuid
GROUP BY i.name
ORDER BY SUM(gd.a_count)
LIMIT 1
```

Table 4: 1 records

name
ABDUL KHAN

Qual é a disciplina mais difícil?

```
SELECT s.name FROM subjects AS s
INNER JOIN subject_memberships AS sm ON sm.subject_code = s.code
INNER JOIN course_offerings AS co ON co.uuid = sm.course_offering_uuid
INNER JOIN grade_distributions AS gd ON co.uuid = gd.course_offering_uuid
GROUP BY s.name
ORDER BY SUM(f_count) DESC
LIMIT 1
```

Table 5: 1 records

name
Mathematics

Qual é a disciplina mais fácil?

```

SELECT s.name FROM subjects AS s
INNER JOIN subject_memberships AS sm ON sm.subject_code = s.code
INNER JOIN course_offerings AS co ON co.uuid = sm.course_offering_uuid
INNER JOIN grade_distributions AS gd ON co.uuid = gd.course_offering_uuid
GROUP BY s.name
ORDER BY SUM(a_count) DESC
LIMIT 1

```

Table 6: 1 records

<u>name</u>
Music

4. Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-23 16:34:30 -03"
```